Paths versus proper double comets for the Randić index times algebraic connectivity

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Abstract

For each $n \in \{10, 11, 12, 13\}$, we certify that the path P_n has a strictly smaller product of Randić index and algebraic connectivity than every double comet whose two end stars each have at least two leaves. At order 14, the balanced double comet with two leaves at each end has a smaller product than the path. Two separate exact-arithmetic programs verify these finite comparisons, one by spectral intervals and the other by integer determinant signs. The comparisons contradict the product clause of the 2007 Aouchiche–Hansen–Zheng conjecture under the convention that double comets exclude paths. The printed definition does not explicitly impose this restriction; the inclusive convention is therefore distinguished.

1 The product and the source conjecture

All graphs are finite, simple and connected. For a graph G with vertex degrees d_v, let

$$R(G) = \sum_{uv \in E(G)} \frac{1}{\sqrt{d_u d_v}}, \qquad a(G) = \lambda_2(D - A), \qquad \rho(G) = R(G)a(G),$$

where the Laplacian eigenvalues are ordered increasingly. Aouchiche, Hansen and Zheng [1, p. 98, Conjecture 4] proposed that the minimum of ρ among connected graphs of order n is attained by a path for $n \leq 9$, and by a balanced double comet for $n \geq 10$. Li, Shi and Wang [2, p. 845, Conjecture 1.2] repeated the product clause. Their results on the product impose additional diameter or minimum-degree hypotheses.

Let $\mathrm{DC}(k_1, k_2, p)$ be the tree formed from two centres joined by a path with p internal vertices, with k_1 pendant leaves at the first centre and k_2 at the second. Thus its order is $k_1 + k_2 + p + 2$. Throughout the theorem, a *proper double comet* means $k_1, k_2 \geq 2$ and $p \geq 0$. Balanced double comets with equal star sizes are a subfamily.

The parenthetical definitions in [1, 2] specify equal-sized stars joined by a path, but no lower bound on their size. If one-leaf stars are allowed, $\mathrm{DC}(1, 1, n-4)$ is the path itself. The split into two ranges of n does not logically exclude this interpretation. A later convention in Aouchiche, Hansen, Hertz and Sellal [3, Section 2] restricts balanced double comets to diameter at most $n-3$, which excludes paths, but does not determine the intended scope of the earlier statement.

2 A certified finite comparison

Theorem 1. *For $n \in \{10, 11, 12, 13\}$ and all integers $k_1, k_2 \geq 2$, $p \geq 0$ with $k_1 + k_2 + p + 2 = n$,*

$$\rho(P_n) < c_n < \rho(\mathrm{DC}(k_1, k_2, p)),$$

where the rational constants c_n are given in Table 1. Moreover,

$$\rho(\mathrm{DC}(2, 2, 8)) < \frac{173}{500} < \rho(P_{14}).$$

n	c_n	Unordered classes	Ordered triples
10	49/100	9	15
11	11/25	12	21
12	81/200	16	28
13	373/1000	20	36

Table 1: Exact separating constants. Unordered means $k_1 \leq k_2$; interchanging the two centres gives an isomorphic graph.

Lemma 1. *Let L be the Laplacian of a connected graph of order n, and let J be the all-ones matrix. For $0 < t < n$, positive leading principal minors of $L + J - tI$ imply $a(G) > t$. A negative determinant of $L + J - tI$ implies $a(G) < t$.*

Proof. The all-ones vector is a zero eigenvector of L. On its orthogonal complement, J vanishes. The eigenvalues of $L + J - tI$ are consequently $n - t, \lambda_2(L) - t, \ldots, \lambda_n(L) - t$. Sylvester's criterion proves the first assertion. For the second, a negative determinant forces a negative eigenvalue, and $n - t$ is positive. □

Computer-assisted proof of Theorem 1. The program `verify_principal_minors.py` in the accompanying repository [5] implements the following finite certificate using integer and rational arithmetic only.

For each edge, put $m = d_u d_v$, $s = 10^{12}$, and $q = \lfloor s\sqrt{m} \rfloor$, computed as an integer square root of ms^2. Then

$$\frac{s}{q+1} \leq \frac{1}{\sqrt{m}} \leq \frac{s}{q}.$$

The program also checks these inequalities by squaring the positive rational endpoints. Summing gives positive bounds $R_- \leq R(G) \leq R_+$. To prove $\rho(G) > c$, set $t = c/R_-$ and check the first criterion of Lemma 1. To prove $\rho(G) < c$, set $t = c/R_+$ and check the second. Both uses explicitly check $0 < t < n$.

For $t = a/b$ with $b > 0$, the integer matrix

$$M = b(L + J) - aI$$

has the same principal-minor signs as $L + J - tI$. Each required determinant is computed by fraction-free Bareiss elimination, with row swaps and exact-division checks. For each order in Table 1, the program constructs P_n and enumerates all $k_1 \leq k_2$ with $k_1, k_2 \geq 2$ and $k_1 + k_2 + 2 \leq n$, setting $p = n - k_1 - k_2 - 2$. Every path gives the required negative determinant at c_n/R_+, and

every proper double comet gives positive leading principal minors at c_n/R_-. At order 14 the path and $\mathrm{DC}(2,2,8)$ satisfy the opposite comparisons at 173/500.

These are 63 product comparisons: 57 proper double comets, four paths, and two order-14 comparisons. The program checks the enumeration counts, graph simplicity, connectedness, degree multisets and edge counts, and records each rational spectral threshold. Every comparison succeeds. Its checks use explicit exceptions and remain active in optimized Python. □

A second program, `verify_theorem.py`, independently bounds each algebraic connectivity in two ways: rational inertia counts for $L - tI$, and Sturm isolation for the characteristic polynomial after removing its simple zero factor. It computes characteristic polynomials by the exact Faddeev–LeVerrier recurrence and checks Cayley–Hamilton. For paths it also checks

$$R(P_n) = \frac{n-3}{2} + \sqrt{2}, \qquad a(P_n) = 2 - 2\cos(\pi/n)$$

using rational bounds for the radicals, π and cosine. Optional SymPy characteristic polynomials provide a further implementation check. Floating point supplies trial endpoints and a noncertifying closed-form sanity check; all final product comparisons use exact rational bounds.

The interval program checks both orders of the star sizes. It also certifies that $\mathrm{DC}(2,2,n-6)$ is the unique smallest proper-double-comet class for each $n = 10, \ldots, 14$, by strict separation of its upper product bound from every other class's lower bound. At order 14 all other proper-double-comet classes are still above the path.

3 Interpretation and supplementary computation

Under the proper convention, Theorem 1 excludes every balanced double comet from being a global minimizer at orders 10–13: the path is a connected graph of the same order with a smaller value. Proving that the path is globally optimal is unnecessary for this conclusion. Under the inclusive convention, the theorem does not refute the conjecture, because the path belongs to the named family. The order-14 comparison does not establish a threshold valid for all larger orders.

Separate saved floating-point scans report P_{10} as the minimizer among 11,716,571 connected graphs of order 10. Tree scans through order 22 report paths for orders 10–13, $\mathrm{DC}(2,2,n-6)$ for 14–17, $\mathrm{DC}(3,3,n-8)$ for 18–21, and $\mathrm{DC}(4,4,12)$ for 22. These are numerical observations, not exact global optimality certificates. The repository includes a replay of 40 retained winners and runners-up, not a rerun of the full enumeration. The interrupted connected-graph scan at order 11 is excluded.

Gebendorfer's tadpole construction [4] concerns the quotient R/a, a separate clause. It should not be conflated with these product comparisons. The accompanying literature record identifies the sources searched; no earlier instance of this precise finite comparison was located, and no exhaustive historical-priority claim is made.

Reproducibility

The exact verifiers require Python 3.10 or later and no external packages:

```
python3 -I certificate/verify_theorem.py --no-sympy
python3 -I -O certificate/verify_principal_minors.py
```

The first command ends with `ALL CERTIFIED: True`; the second outputs a JSON record with `all_certified` equal to `true`. Both return a nonzero exit status on a failed check. Source code, replay outputs, review provenance, and this note are available in [5]. The note is licensed under Creative Commons Attribution 4.0 International.

References

[1] M. Aouchiche, P. Hansen, and M. Zheng, Variable Neighborhood Search for Extremal Graphs. 19. Further Conjectures and Results about the Randić Index, *MATCH Commun. Math. Comput. Chem.* **58** (2007), 83–102. Publisher PDF.

[2] X. Li, Y. Shi, and L. Wang, On a Relation Between Randić Index and Algebraic Connectivity, *MATCH Commun. Math. Comput. Chem.* **68** (2012), 843–849. Publisher PDF.

[3] M. Aouchiche, P. Hansen, A. Hertz, and C. Sellal, *Bounds on differences between some graph theoretic invariants*, Les Cahiers du GERAD G–2015–31, April 2015. Report.

[4] J. J. Gebendorfer, *An infinite tadpole counterexample family to the Randić index–algebraic connectivity conjecture*, preprint, August 6, 2026. doi:10.5281/zenodo.21827918.

[5] J. Prajapati, *Paths versus proper double comets for Randić index times algebraic connectivity*, computational artifact, version 1.0.0, November 5, 2026. `https://github.com/infinityscroll/agx-randic-connectivity-certificate`.